HIGH-EFFICIENCY TEST ARCHITECTURE FOR 3D TORUS NoC USING XOR-BASED CODING

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IN
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CERTIFICATE

This is to certify that the project report entitled "**High-Efficiency Test Architecture for 3D Torus NoC Using XOR-Based Coding**" being submitted by **Madasu Teja Sai Vivek (23KT5A0405), Bevara Raghunadh (22KT1A0434), Bayana Naveen (22KT1A0433), and Alapati Karthik Sai (22KT1A0431)** in partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY to the Jawaharlal Nehru Technological University, Kakinada is a record of Bonafide work carried out under my guidance and supervision. The results embodied in this project report have not been submitted to any other university or institution for the award of any degree or diploma.

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We certify that the work contained in this report is original and has been done by us under the guidance of my supervisor. The work has not been submitted to any other Institute for any degree or diploma. We have followed the guidelines provided by the Institute in preparing the report. we have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

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Program Educational Objectives (PEOs) of the Department

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PEO 2 (Core Competence):	To motivate graduates to Analyze, Design, Develop, Optimize and Implement electronic systems with competent spirit.
PEO3 (Breadth Knowledge):	To enable graduates with sufficient breadth in electronics and its related fields to solve general engineering problems in an eco-friendly environment.
PEO 4 (Soft Skills):	To make graduates with a professional outlook where it can communicate effectively and interact responsibly with colleagues, clients, employers and society.
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IV Year II Semester Project Course Outcomes

CO's	Course Outcomes	BTL
CO1	Identify the complex engineering problems relevant to the society and industry	Apply
CO2	Apply modern technologies, tools, and systems in the field of Electronics & Communication Engineering to analyze the identified problem	Apply
CO3	Design and implement a viable solution to the problem	Create
CO4	Apply Communication & Presentation skills	Apply
CO5	Develop the team work, leadership skills with professional and ethical values.	Understand
CO6	Make the use of report writing skills	Understand

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ABSTRACT

Complexity grows in multi-core setups and IoT uses, pushing more designs toward 3 -D Network-on-Chip systems for better internal data flow. Testing router functions matters a lot for stable performance; however old ways take too long and slow things down - especially when chips get bigger and stack in 3-D grids. To fix this, a smarter test approach using network coding is introduced here, tailored for 3 -D Torus NoCs. Instead of standard grid layouts, the Torus shape loops connections at edges, cutting how far signals must travel. Because of those loops, delays drop sharply across the chip. Routing shifts from basic Dimension-Order rules to a modulo-style system, which finds tighter routes while spreading traffic more evenly. Paths shrink. Bottlenecks fade. Efficiency rises without extra hardware. Instead of sending each packet alone, the new setup uses XOR coding inside routers to bundle several at once. This bundling cuts down how many routes are needed during checks, which shortens total test duration. Tests run through simulation show lower delays and better data flow rates than older mesh techniques. Hardware tests on FPGA boards confirm quicker error spotting alongside these gains. The design fits well within modern chip frameworks where power saving matters. Features like clock gating and power gating help reduce energy use across the network layers. Work happens directly on programmable chips to validate real-world function. Efficiency improvements appear without increasing complexity too much. Ideas apply mainly to systems built as single integrated units with shared communication backbones.

Keywords: Network-on-Chip (NoC), XOR Network Coding, Modulo Dimension Routing (MDR), FPGA Implementation.

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LIST OF ABBREVIATIONS

Abbreviation	Description
NoC	Network-on Chip
3-D NoC	Three-Dimensional Network-on-Chip
SoC	System-on-Chip
MPSoC	Multi-Processor System-on-Chip
TSV	Through-Silicon Via
MDR	Modulo Dimension Routing
XOR-NC	XOR-based Network Coding
ECC	Error Correction Code
BIST	Built-In Self-Test
FIFO	First In First Out
FPGA	Field Programmable Gate Array
HDL	Hardware Description Language
VLSI	Very Large Scale Integration
IoT	Internet of Things
RF	Radio Frequency
VSWR	Voltage Standing Wave Ratio
S11	Return Loss Parameter
S21	Insertion Loss Parameter

CHAPTER 1

INTRODUCTION

1.1 Overview of Modern Computing and NoC

The rapid advancement in semiconductor technology has revolutionized the design and functionality of modern computing systems. With continuous scaling of transistor sizes and the integration of billions of transistors on a single chip, systems have evolved from simple single-core processors to highly complex multi-core and many-core architectures. These modern systems support a wide range of applications such as artificial intelligence, machine learning, cloud computing, high-performance computing, and Internet of Things devices. Such applications require not only massive computational power but also efficient and high-speed data communication between processing elements.

In earlier computing architectures, communication between components such as processors and memory was handled using shared bus-based systems. While these bus architectures were simple and cost-effective for small-scale systems, they suffer from significant limitations when scaled to larger systems. The shared communication medium becomes a bottleneck due to limited bandwidth, increased contention among components, higher latency, and reduced overall system performance. As the number of cores increases, these limitations become more severe, making traditional communication methods unsuitable for modern computing requirements.

To overcome these issues, Network-on-Chip (NoC) has emerged as a structured and scalable communication paradigm. In NoC architecture, the traditional bus is replaced by a network-like structure where multiple components such as processors, memory units, and accelerators communicate through routers and interconnecting links. Each processing element is connected to a router, and routers are interconnected in a predefined topology. Data is transmitted in the form of packets containing source and destination information along with payload data. This packet-based communication enables multiple data transfers to occur simultaneously across different paths, which improves throughput, reduces congestion, and enhances overall system

performance. As a result, NoC provides better scalability, supports parallel communication, reduces latency, improves bandwidth utilization, and enables modular system design. Due to these advantages, NoC has become a fundamental component in modern System-on-Chip design.

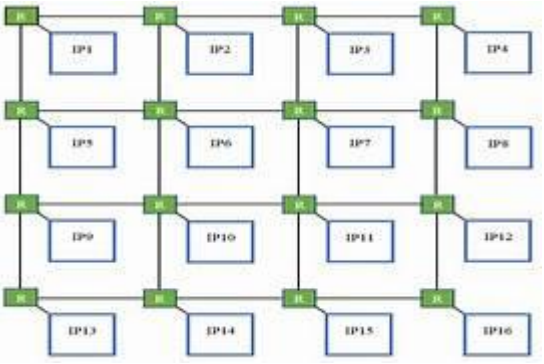


Figure 1.1: 4x4 2D Mesh NoC

1.2 Router Functionality and Testing Challenges

Routers are the core components of a NoC system and play a crucial role in managing communication between different nodes. Each router is responsible for receiving data packets from its input ports, temporarily storing them in buffers, making routing decisions, resolving conflicts through arbitration, allocating internal switching paths, and forwarding packets to the appropriate output ports. These operations ensure that data is correctly delivered from source to destination across the network.

Since routers control the flow of data, any malfunction in their operation can lead to serious issues such as data corruption, packet loss, increased latency, network congestion, or even complete system failure. Therefore, ensuring the correct functionality of routers is critical for the overall reliability and performance of the NoC system. Functional testing of routers is an essential process that verifies whether routers operate correctly under different traffic conditions and communication scenarios.

However, testing routers in NoC systems presents several challenges. As the network size increases, the complexity of testing also grows significantly. A large number of communication paths must be verified to ensure complete coverage, which increases testing time and computational effort. Additionally, multiple packets may compete for the same communication resources, making it necessary to test arbitration and conflict resolution mechanisms. Traditional testing approaches often require extensive hardware resources and are not scalable for large systems. As a result, there is a need for more efficient testing techniques that can reduce testing time, minimize resource usage, and maintain high accuracy in fault detection.

1.3 Transition to 3-D NoC and Limitations of Mesh Topology

With continuous advancements in technology, NoC architectures have evolved from two-dimensional designs to three-dimensional structures. In 3-D NoC, multiple layers of circuits are stacked vertically, allowing communication not only in horizontal directions but also in the vertical dimension. This is typically achieved using technologies such as Through-Silicon Vias, which enable direct connections between different layers of the chip. The introduction of the third dimension significantly reduces interconnect length, increases bandwidth, lowers power consumption, and improves overall system performance. It also allows higher integration density, enabling more components to be placed within a smaller area.

Despite these advantages, 3-D NoC introduces additional complexity in routing, design, and testing. The most commonly used topology in such systems is the mesh topology due to its simple and regular structure. However, mesh topology suffers from several limitations. In a mesh network, routers located at the edges and corners have fewer connections compared to those in the centre, resulting in unequal connectivity and inefficient utilization of network resources. This imbalance can lead to uneven traffic distribution and increased congestion in certain areas of the network.

Another limitation of mesh topology is its relatively large network diameter, which refers to the maximum distance between any two nodes. A larger diameter results in longer communication paths, thereby increasing latency. Additionally, the absence of wrap-around links means that nodes at opposite edges of the network are not directly connected, forcing packets to travel longer distances. This further degrades performance. From a testing

perspective, mesh topology is inefficient because it requires independent verification of multiple communication paths across different layers. As the size of the network increases, the number of required test operations grows rapidly, making the testing process time-consuming and less scalable. These limitations highlight the need for an improved topology that can provide better performance and more efficient testing.

1.4 Proposed 3-D Torus Architecture with Efficient Techniques

To overcome the limitations of mesh topology, this work proposes a 3-D Torus NoC architecture. The torus topology enhances the mesh structure by introducing wrap-around links in all three dimensions, namely X, Y, and Z. These links connect the nodes at opposite edges of the network, effectively eliminating boundary conditions and ensuring uniform connectivity across all routers. As a result, each router has the same number of connections, which leads to balanced traffic distribution and improved network efficiency.

The addition of wrap-around links reduces the network diameter, allowing packets to travel through shorter cyclic paths. This significantly improves communication speed and reduces latency. Furthermore, the torus topology provides multiple path options for data transmission, enabling packets to take alternative routes in case of congestion or faults. This enhances fault tolerance and overall system reliability.

To further optimize routing performance, Modulo Dimension Routing (MDR) is employed. This routing algorithm determines the shortest path by considering wrap-around distances in each dimension. By selecting the minimum distance between source and destination, MDR reduces the number of hops required for packet transmission, thereby decreasing latency and improving efficiency.

In addition to topology and routing improvements, the proposed work incorporates XOR-based network coding for functional testing. In this approach, multiple test packets are combined using XOR operations and transmitted as a single packet. At the destination, the original packets are reconstructed by applying the XOR operation again. This method reduces the number of transmissions required for testing, thereby

decreasing testing time and resource consumption. At the same time, it ensures that the original data can be accurately recovered, maintaining correctness and reliability. This innovative testing technique significantly improves scalability and efficiency compared to traditional methods.

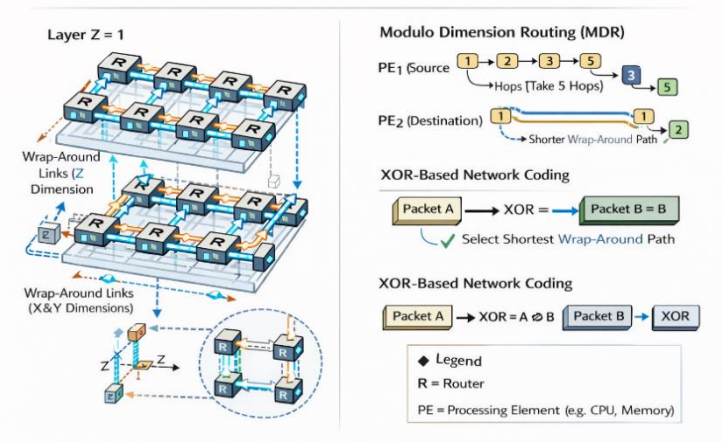


Figure 1.2: Proposed 3D Torus NoC

1.5 Objectives, Scope, and Contribution

The main objective of this work is to design an efficient and scalable NoC architecture that enhances both communication performance and testing efficiency. This is achieved by transforming the traditional 3-D mesh topology into a 3-D torus, implementing Modulo Dimension Routing for optimal path selection, and applying XOR-based network coding to reduce testing complexity. The proposed approach aims to minimize latency, improve throughput, and ensure reliable data transmission while reducing hardware overhead.

The scope of this work includes the design and implementation of the 3-D Torus topology, development of an efficient routing mechanism, and incorporation of advanced testing techniques. The system is modelled and verified using Verilog HDL, which allows simulation and analysis of the proposed architecture. Performance evaluation is carried out based on parameters such as latency, throughput, network diameter, and resource utilization. The focus is on achieving high performance and scalability while maintaining low power consumption and efficient resource usage.

The key contribution of this work lies in integrating topology optimization, efficient routing, and coding-based testing into a single unified framework. Unlike existing approaches that address these aspects independently, the proposed method combines them to provide a comprehensive solution for modern NoC systems. The conversion of mesh topology to torus improves connectivity and reduces latency, while MDR enhances routing efficiency. The use of XOR-based network coding significantly reduces testing complexity and time. This integrated approach makes the proposed architecture highly suitable for next-generation computing systems that require fast communication, reliable operation, and scalable design.

1.6 Organization of Thesis

The thesis is systematically structured into seven chapters, routing algorithms, functional testing, and system implementation using Xilinx tools. It also evaluates the performance of the proposed architecture in terms of latency, throughput, and reliability, demonstrating its effectiveness for modern multi-core and many-core systems.

Chapter 1: Introduction

This chapter introduces the need for efficient communication systems in modern multi-core and many-core architectures. Traditional methods such as shared buses and point-to-point communication face limitations like high latency and poor scalability. To overcome these issues, Network-on-Chip (NoC) has been developed as an efficient communication framework. This chapter also highlights the motivation for using 3-D NoC architectures and outlines the objectives of the proposed 3-D Torus NoC system.

Chapter 2: Literature Survey

This chapter presents a review of existing research related to NoC systems. It discusses different topologies such as mesh and torus, routing algorithms like Modulo Dimension Routing (MDR), and testing techniques including network coding. The advantages and limitations of previous works are analysed, and the research gap is identified. This

provides the foundation for developing the proposed system.

Chapter 3: NoC Topology

This chapter explains the concept of NoC topology and its importance in system performance. Different types of topologies such as mesh and torus are discussed in detail. The limitations of mesh topology and the advantages of torus topology, such as reduced latency and uniform connectivity, are highlighted. The structure of 3-D NoC topology and the role of wrap-around links are also explained.

Chapter 4: Xilinx Software

This chapter provides an overview of Xilinx software used for the design and simulation of the proposed system. It explains the features of Xilinx Vivado, including design entry, synthesis, simulation, and implementation. The process of writing HDL code, running simulations, and analysing outputs is also described. This chapter helps in understanding how the proposed NoC design is implemented and verified.

Chapter 5: Design and Implementation of 3-D Torus NoC

This chapter describes the proposed 3-D Torus NoC architecture in detail. It explains how the 3-D mesh topology is converted into a torus structure using wrap-around links. The design of routers, communication links, and routing mechanism using MDR is presented. The implementation of XOR-based network coding for efficient testing is also discussed, along with the overall working of the system.

Chapter 6: Results and Discussion

This chapter presents the simulation results obtained from the implementation of the proposed system. The performance is evaluated based on parameters such as latency, throughput, and reliability. The results are compared with traditional mesh-based NoC

systems to show the improvements achieved by the 3-D torus topology and efficient routing. The observations and analysis are discussed clearly.

Chapter 7: Conclusion and Future Scope

This chapter summarizes the overall work and highlights the key contributions of the proposed system. It concludes that the 3-D Torus NoC provides improved performance, reduced latency, and better scalability compared to existing systems. The future scope includes enhancements such as adaptive routing, thermal management, and integration of advanced technologies to further improve system efficiency.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The rapid advancement of semiconductor technology and continuous scaling of integrated circuits have enabled the development of highly complex multi-core and many-core systems. These systems requires efficient and high-speed communication mechanisms to ensure proper coordination among processing elements. Traditional communication architectures such as shared buses andThe rapid advancement of semiconductor technology and the continuous scaling of integrated circuits have led to the development of highly complex multi-core and many-core systems. Modern System-on-Chip (SoC) designs integrate hundreds of processing elements, memory units, and specialized accelerators on a single chip. As the number of cores increases, the need for efficient and high-speed communication among these components becomes critical for overall system performance. Traditional communication architectures such as shared buses and point-to-point interconnects are no longer suitable due to their inherent limitations in scalability, latency, and bandwidth.

In earlier systems, shared bus architectures were widely used due to their simplicity and ease of implementation. However, as the number of processing elements increases, bus-based systems suffer from contention and bandwidth bottlenecks. Only one communication transaction can occur at a time, which significantly limits parallelism and reduces system throughput. Similarly, point-to-point interconnects, although offering dedicated communication paths, become highly complex and inefficient when scaled to large systems. The number of links grows rapidly with the number of nodes, leading to increased area, power consumption, and design complexity.

To overcome these challenges, Network-on-Chip (NoC) has emerged as a promising communication paradigm. NoC replaces traditional interconnects with a network-based approach, where communication is carried out using packets transmitted through routers and links. This approach enables parallel data transmission, improves bandwidth utilization, and significantly enhance system scalability. In a typical NoC architecture, each processing element is connected to a router, and routers are interconnected using a specific topology such as mesh, torus, or tree.

One of the key advantages of NoC is its ability to support scalable communication. As the system grows, additional routers and links can be added without significantly affecting performance. Regular topologies like 2D mesh are widely used due to their simplicity and ease of implementation. However, advanced topologies such as 3D mesh and 3D torus have gained attention for their ability to further reduce communication latency and improve network efficiency by introducing additional connectivity in the third dimension.

Routing algorithms play a crucial role in determining the performance of NoC systems. Efficient routing ensures that data packets travel from source to destination with minimal delay and congestion. Deterministic routing algorithms, such as XY routing, are simple and deadlock-free but may not always provide optimal performance under heavy traffic conditions. On the other hand, adaptive routing algorithms dynamically select paths based on network conditions, helping to balance load and reduce congestion. However, these methods introduce additional complexity in router design.

Another important aspect of NoC design is flow control, which manages the movement of packets within the network. Techniques such as wormhole switching and virtual channel flow control are commonly used to improve performance and avoid deadlocks. Wormhole switching divides packets into smaller units called flits, allowing efficient utilization of network resources. Virtual channels help in reducing blocking and improving throughput by providing multiple logical paths within a single physical channel.

Power consumption is also a critical concern in NoC design. As the number of routers and links increases, the overall power consumption of the network becomes significant. Researchers have proposed various techniques to address this issue, including power-aware routing, dynamic voltage scaling, and energy-efficient router architectures. Minimizing power consumption while maintaining high performance remains a key challenge in NoC systems.

In addition to power and performance, reliability is another important factor in modern NoC architectures. With technology scaling, the probability of faults increases due to variations, aging, and environmental factors. Fault-tolerant routing algorithms and error detection mechanisms are employed to ensure reliable communication in the presence

of faults. Redundant paths and reconfigurable architectures further enhance system robustness.

Recent research has also focused on 3D Network-on-Chip architectures, which utilize vertical interconnections such as Through-Silicon Vias (TSVs) to connect multiple layers of chips. 3D NoC offers significant advantages in terms of reduced interconnect length, lower latency, and improved bandwidth. Among these, 3D torus topology provides wrap-around connections in all three dimensions, eliminating edge nodes and ensuring uniform communication delay across the network. This makes it highly efficient for large-scale systems.

Furthermore, simulation tools such as Book Sim and other NoC simulators are widely used to evaluate network performance under different traffic patterns and configurations. These tools help researchers analyse metrics such as latency, throughput, power consumption, and packet delivery ratio. By comparing different topologies and routing algorithms, optimal designs can be identified for specific applications.

In summary, Network-on-Chip has become a fundamental solution for addressing communication challenges in modern multi-core and many-core systems. Its ability to provide scalable, high-performance, and energy-efficient communication makes it a preferred choice over traditional interconnect architectures. Ongoing research continues to focus on improving routing efficiency, reducing power consumption, enhancing fault tolerance, and exploring advanced topologies such as 3D torus networks.

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CHAPTER 3

NETWORK-ON-CHIP TOPOLOGY

3.1 Introduction to NoC Topology

In modern System-on-Chip (SoC) and Multi-Processor System-on-Chip (MPSoC) designs, efficient communication between multiple processing elements is a fundamental requirement. As the number of cores increases, the communication architecture must handle high data traffic with minimal delay, power consumption, and congestion. Network-on-Chip (NoC) has emerged as a scalable and structured solution to address these challenges.

A key aspect of NoC design is topology, which defines how routers (nodes) are interconnected through communication links. The topology determines the overall structure of the network and has a direct impact on critical performance parameters such as latency, throughput, power consumption, scalability, and fault tolerance.

In NoC architectures, data is transmitted in the form of packets through routers using predefined routing algorithms. The efficiency of packet delivery largely depends on how well the topology supports multiple communication paths, minimizes hop count, and balances network load. A well-designed topology ensures uniform connectivity, reduces bottlenecks, and improves overall system performance.

With advancements in chip fabrication, NoC topologies have evolved from simple two-dimensional (2-D) structures to more complex three-dimensional (3-D) architectures. These 3-D topologies provide additional vertical connections, enabling shorter communication paths and improved bandwidth. However, they also introduce challenges such as increased design complexity and testing overhead.

Therefore, selecting an appropriate topology is crucial for achieving efficient communication and reliable operation in NoC systems.

3.2 Types of NoC Topologies

NoC topologies can be broadly classified into several categories based on their structure and connectivity. Each topology has its own advantages and limitations.

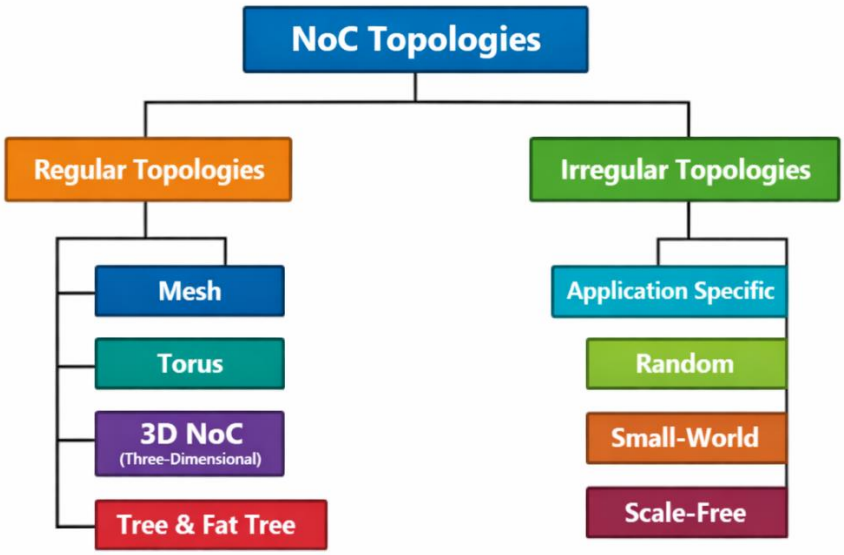


Figure 3.1: NoC topologies family tree

1 Bus Topology

In bus topology, all nodes are connected to a single shared communication channel.

Features:

Simple design and easy implementation

Low hardware cost

Limitations:

High congestion due to shared medium

Poor scalability

Single point of failure

2 Ring Topology

In ring topology, each node is connected to two neighbouring nodes, forming a circular structure.

Features:

Simple and structured design

Equal load distribution

Limitations:

High latency for long-distance communication

Failure of one node can affect the entire network

3 Star Topology

In star topology, all nodes are connected to a central hub or controller.

Features:

Easy to manage and control

Efficient communication for small systems

Limitations:

Central node becomes a bottleneck

Not scalable for large NoCs

4 Mesh Topology (2-D Mesh)

Mesh topology is one of the most widely used NoC structures. Nodes are arranged in a grid format, and each node is connected to its neighboring nodes.

Features:

Regular and simple structure

Easy routing (e.g., XY routing)

Limitations:

Edge and corner nodes have fewer connections

Higher latency for distant nodes

Unequal load distribution

5 Tree Topology

Tree topology organizes nodes in a hierarchical structure.

Features:

Efficient for hierarchical communication

Easy data aggregation

Limitations:

Root node failure affects the entire system

Limited scalability

6 Torus Topology

Torus topology is an extension of mesh topology where opposite edges are connected using wrap-around links.

Features:

Uniform connectivity for all nodes

Reduced network diameter

Limitations:

Slightly increased hardware complexity

Additional wrap-around links required

7 3-D NoC Topology

In 3-D NoC, multiple layers of 2-D networks are stacked vertically.

Features:

Shorter communication paths

Higher bandwidth

Better integration density

Limitations:

Complex routing

3.3 Main Topology Used: 3-D Torus Topology

The proposed work uses a 3-D Torus topology, which is an advanced extension of both mesh and torus structures.

3.3.1 Structure of 3-D Torus

In a 3-D Torus:

Nodes are arranged in a 3-dimensional grid (X, Y, Z axes)

Each node is connected in six directions:

East, West

North, South

Up, Down

Wrap-around links connect opposite edges in all three dimensions

This creates a cyclic structure, eliminating boundaries present in mesh topology.

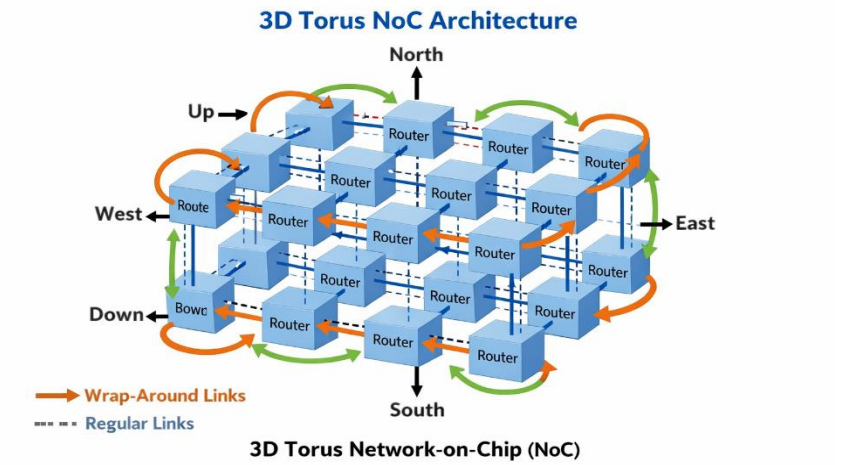


Fig 3.3.1: 3D Torus Network on Chip

3.3.2 Advantages of 3-D Torus

1. Uniform Connectivity

All routers have the same number of connections, unlike mesh where edge nodes have fewer links.

2. Reduced Network Diameter

Wrap-around links allow shorter paths, reducing hop count and latency.

3. Improved Load Balancing

Traffic is distributed evenly across the network.

4. Better Scalability

Efficient for large-scale systems due to reduced path length.

5. Efficient Routing

Supports advanced routing algorithms like Modulo Dimension Routing (MDR).

3.3.3 Role in Proposed Work

In the given project:

The 3-D Mesh topology is converted into a 3-D Torus

Wrap-around links are added in X, Y, and Z directions

This improves:

Testing efficiency

Routing performance

Delay reduction

Additionally, the torus topology works effectively with XOR-based network coding, enabling:

Reduced number of test paths

Faster functional testing

Better resource utilization

3.3.4 Comparison with 3-D Mesh

Feature	**3-D Mesh**	**3-D Torus**
Connectivity	Unequal (edges weaker)	Uniform
Network Diameter	High	Low
Path Length	Longer	Shorter
Load Distribution	Uneven	Balanced

Table 1: comparison between 3D Mesh and 3D Torus

3.3.5 Comparison of NoC Topologies

Below is a detailed comparison of major NoC topologies:

Topology	Bus	Mesh	Tree	Torus	3-D Mesh
Structure	Shared channel	Grid	Hierarchical	Mesh + wrap	Layered grid
Scalability	Low	High	Medium	High	High
Latency	High	Medium	Medium	Low	Medium
Complexity	Low	Medium	Medium	Medium	High
Advantages	Simple, low cost	Simple routing	Structured communication	Short paths, balanced	Better bandwidth
Limitations	Congestion, not scalable	Edge node limitation	Root dependency	Extra links	Complex routing

Table 2: Comparison between different types of topologies

CHAPTER 4

SOFTWARE TOOLS

4.1 Introduction to Xilinx Software

Xilinx software is a comprehensive development environment used for designing, simulating, and implementing digital systems on Field Programmable Gate Arrays (FPGAs). It provides a complete platform that supports hardware design using Hardware Description Languages (HDL) such as Verilog and VHDL. These tools are widely used in both academic research and industrial applications for building complex digital architectures, including processors, communication systems, and Network-on-Chip (NoC) designs.

Among the available tools, Xilinx Vivado is the most advanced and widely used design suite, offering improved synthesis capabilities, faster simulation, and efficient debugging features. Another tool, Xilinx ISE, is an earlier generation software that is still used for older FPGA devices but has been largely replaced by Vivado.

In the context of NoC design, Xilinx software plays a crucial role in modelling and verifying router architectures. Designers can create modules representing routers, buffers, arbiters, and routing logic, and then simulate their behaviour under different conditions. The software ensures that the design functions correctly before it is implemented on actual hardware. This reduces development time and minimizes errors, making the design process more efficient and reliable.



Fig 4.1: Xilinx Vivado

4.2. Design Flow in Xilinx

The design flow in Xilinx software follows a structured approach that transforms a high-level hardware description into a physical implementation on an FPGA. This flow ensures that the design is accurate, optimized, and ready for real-time operation.

The process begins with design entry, where the designer writes the HDL code describing the functionality of the system. In NoC projects, this includes defining router behaviour, packet handling mechanisms, and routing algorithms. Each module is carefully designed to ensure proper communication between nodes. After design entry, functional simulation is performed. Simulation is a critical step that allows designers to test the logic of the system without using physical hardware. Test benches are created to apply various input conditions and observe the output responses. This helps in identifying logical errors, verifying packet transmission, and ensuring correct routing decisions.

Once the design is verified, the next step is synthesis. During synthesis, the HDL code is converted into a gate-level representation. The tool maps the design into FPGA resources such as Look-Up Tables (LUTs), flip-flops, and multiplexers. This step also performs optimization to reduce resource usage and improve performance.

Following synthesis, the design undergoes implementation, which includes mapping, placement, and routing. In mapping, logical components are assigned to physical FPGA resources. During placement, these components are positioned on the FPGA chip, and routing establishes the connections between them. This stage determines the physical layout of the design and has a significant impact on timing and performance.

Finally, a bitstream file is generated, which is used to configure the FPGA. The design is then loaded onto the hardware for real-time testing and validation. This complete flow ensures that the system is both functionally correct and physically implementable.

4.3 Parameters Obtained from Xilinx Software

Xilinx software provides detailed analysis reports that help designers evaluate the performance, efficiency, and reliability of their designs. These parameters are essential for understanding how well the system operates and where improvements can be made.

One of the most important categories is timing parameters. These include total delay, logic delay, and net delay. Total delay represents the time taken for a signal to travel from input to output. Logic delay is caused by internal components such as gates and multiplexers, while net delay is due to interconnections between components. Additional timing parameters such as setup time and hold time ensure proper synchronization of signals. The maximum operating frequency (Fmax) indicates the highest speed at which the circuit can function reliably.

Another important category is resource utilization parameters. These parameters indicate how efficiently the FPGA hardware is used. They include the number of LUTs used for implementing combinational logic, flip-flops for storing data, slices as basic building units, block RAM (BRAM) for memory storage, and DSP units for arithmetic operations. Efficient resource utilization is crucial for fitting the design within the available hardware and ensuring cost-effectiveness.

The software also provides power analysis, which includes dynamic power, static power, and total power consumption. Dynamic power is consumed during switching activity, while static power is consumed even when the circuit is idle. Managing power consumption is especially important in modern systems such as IoT devices and embedded applications, where energy efficiency is a key requirement.

In addition, routing and performance parameters are analysed. These include fanout, which represents the number of outputs driven by a signal, and routing congestion, which indicates how heavily the routing resources are utilized. The critical path is the longest delay path in the circuit and determines the overall speed of operation. Performance metrics such as throughput and latency are also evaluated, particularly in NoC designs, where they directly affect communication efficiency.

4.4 Importance of Parameters in the Proposed NoC Design

In the proposed 3-D Torus NoC architecture, the parameters obtained from Xilinx software are used to evaluate and validate the performance improvements over traditional mesh-based systems.

Timing parameters play a key role in demonstrating the efficiency of the design. The use of Modulo Dimension Routing reduces the number of hops required for packet transmission,

which in turn reduces total delay. The presence of wrap-around links in the torus topology allows packets to take shorter paths, improving overall speed and reducing latency.

Resource utilization analysis shows that the additional hardware required for implementing XOR-based network coding introduces only a minimal increase in resource usage. This ensures that the design remains efficient and practical for implementation on FPGA devices. The balance between performance improvement and hardware cost is an important factor in evaluating the effectiveness of the proposed architecture.

Power analysis indicates that reducing the number of packet transmissions through XOR coding can lead to lower dynamic power consumption. This contributes to energy efficiency, which is an important consideration in modern electronic systems. Routing and performance parameters further highlight the advantages of the torus topology. Uniform connectivity ensures that all routers have equal communication opportunities, reducing congestion and improving load balancing. The critical path delay is minimized due to shorter routing paths, leading to better overall system performance. These results confirm that the proposed 3-D Torus NoC architecture provides significant improvements in delay, efficiency, and scalability compared to traditional mesh-based designs.

Xilinx software provides a powerful and complete platform for designing and analysing digital systems, particularly those implemented on FPGAs. It supports every stage of the design process, from initial coding and simulation to final hardware implementation and testing. The parameters generated by Xilinx tools, including timing analysis, resource utilization, power consumption, and routing performance, provide valuable insights into the behaviour of the system. These parameters help designers optimize their designs and ensure efficient operation. In the context of the proposed 3-D Torus NoC architecture, Xilinx software plays a crucial role in validating the design and demonstrating its advantages. The results show improved delay performance, efficient resource usage, and better scalability, confirming that the proposed system is suitable for modern high-performance applications. Overall, Xilinx tools are essential for developing reliable and efficient digital systems, making them a key component in the design of advanced NoC architectures.

CHAPTER 5

DESIGN AND IMPLEMENTATION OF 3-D TORUS NoC

5.1 Introduction

The rapid advancement in semiconductor technology has enabled the integration of a large number of processing elements onto a single chip, leading to the development of complex System-on-Chip (SoC) architectures. However, as the number of cores increases, efficient communication between these components becomes a critical challenge. Traditional communication architectures such as shared buses and point-to-point interconnects are no longer sufficient due to their limited scalability, increased latency, and high-power consumption.

To overcome these limitations, the concept of Network-on-Chip (NoC) has emerged as a powerful alternative. NoC replaces conventional interconnects with a structured communication network, enabling scalable and efficient data transfer between cores. Among various NoC topologies, the torus topology has gained significant attention due to its ability to provide uniform connectivity and reduced communication delay.

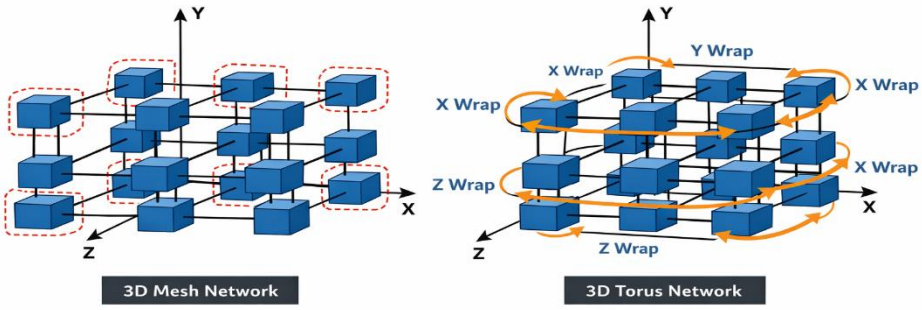


Figure 5.1: Comparison between 3-D Mesh and 3-D Torus Topology

This chapter presents the design and implementation of a **3-D Torus Network-on-Chip (NoC)** using Verilog Hardware Description Language (HDL). The proposed architecture enhances the traditional 3-D mesh topology by introducing wrap-around connections along all three spatial dimensions—X, Y, and Z—forming a toroidal structure. Unlike mesh networks where boundary nodes suffer from reduced connectivity, the torus topology ensures that every node has an equal number of connections, resulting in balanced traffic distribution and improved performance.

The design methodology follows a structured approach, starting from topology transformation, followed by router design, routing algorithm implementation, and optimization using an XOR-based testing mechanism. The complete system is simulated and verified using Xilinx Vivado tools to validate its functional correctness and performance efficiency.

5.2 Architecture Design and Router Implementation

The design of the 3-D Torus NoC begins with the transformation of a conventional 3-D mesh into a torus structure. In a mesh topology, nodes located at the edges have fewer neighboring connections, which leads to uneven communication capability and increased network diameter. This results in longer communication paths and higher latency.

To address these issues, wrap-around connections are introduced. In the X-direction, the nodes on the left edge are connected to those on the right edge. Similarly, in the Y-direction, top nodes connect to bottom nodes, and in the Z-direction, front-layer nodes connect to back-layer nodes. This transformation eliminates boundary conditions and creates a cyclic structure, ensuring uniform connectivity across the network.

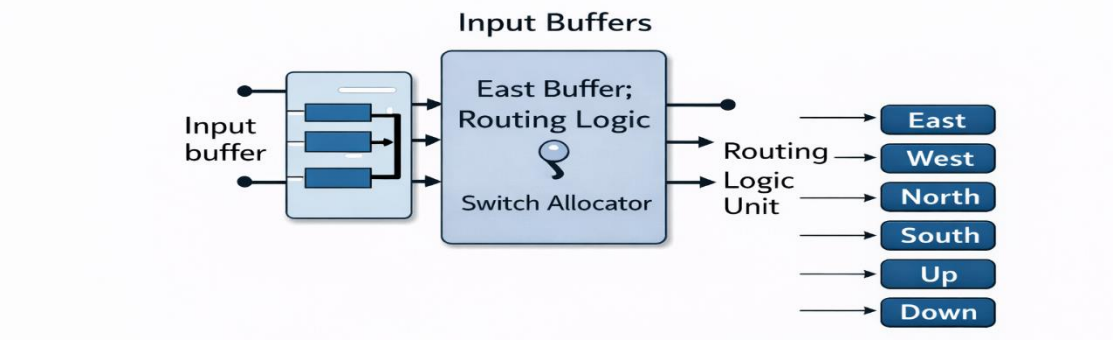


Figure 5.2: Six-Port Router Architecture Block Diagram

Each node in the network is equipped with a router, which acts as the fundamental building block of the NoC. The router is responsible for receiving, storing, processing, and forwarding packets to the appropriate neighboring nodes. The proposed router architecture consists of six ports corresponding to the six possible directions: East, West, North, South, Up, and Down.

The router includes several key components. Input buffers are used to temporarily store incoming packets and prevent data loss during congestion. The routing logic unit determines the optimal path for packet transmission based on the routing algorithm. A switch allocator manages contention when multiple packets request the same output channel, ensuring efficient resource utilization. Finally, output ports forward packets to adjacent routers.

The router is implemented in Verilog HDL, ensuring modularity and scalability. This modular design allows easy expansion of the network without significant redesign effort.

5.3 Routing Algorithm and XOR-Based Optimization

Efficient routing plays a crucial role in determining the overall performance of a NoC system. In this design, the **Modulo Dimension Routing (MDR)** algorithm is employed to ensure optimal packet delivery. MDR is particularly suitable for torus networks as it takes advantage of wrap-around connections to compute the shortest possible path.

The algorithm operates by calculating the distance between the source and destination nodes in each dimension (X, Y, and Z). For each dimension, both forward distance and wrap-around distance are computed using modular arithmetic. The algorithm then selects the shorter of the two distances and routes the packet accordingly, dimension by dimension.

For example, in a 4-node ring, if a packet needs to travel from node 0 to node 3, the forward distance is 3, whereas the wrap-around distance is 1. MDR selects the shorter wrap-around path, thereby reducing latency and improving efficiency.

In addition to efficient routing, the design incorporates an XOR-based testing mechanism to optimize network verification. Traditional testing approaches involve transmitting individual packets, which increases network load and testing time. In contrast, XOR-based network coding combines multiple packets into a single encoded packet using the XOR operation. At the destination, the original packets can be reconstructed using simple XOR decoding.

This approach significantly reduces the number of transmissions required for testing, thereby

improving testing speed and conserving bandwidth. Furthermore, it introduces minimal hardware overhead, making it an efficient solution for large-scale NoC systems.

Algorithm: 3D Torus Routing with Network Coding

Input: Current coordinates (xc, yc, zc) Destination coordinates (xd, yd, zd); Input data D1, D2 with validity flags V1, V2; Network dimensions Nx, Ny, Nz.

Output: Forwarding direction Dir; Output data Dout; Encoding flag f enc; Hop count H.

1.Distance and Hop Count Calculation

Calculate absolute distances:

dx = |xd - xc|, dy = |yd - yc|, dz = |zd - zc|

Calculate shortest torus distances:

dx_t = min(dx, Nx - dx)

dy_t = min(dy, Ny - dy)

dz_t = min(dz, Nz - dz)

H = dx_t + dy_t + dz_t

2.Routing Strategy (Torus DOR)

if xc ≠ xd then

if (xd - xc + Nx) (mod Nx) ≤ (xc - xd + Nx)

(mod Nx) then Dir = EAST

else Dir = WEST

else if yc ≠ yd then

if (yd - yc + Ny) (mod Ny) ≤ (yc - yd + Ny)

(mod Ny) then Dir = NORTH

else Dir = SOUTH

else if zc ≠ zd then

if (zd - zc + Nz) (mod Nz) ≤ (zc - zd + Nz)

(mod Nz) then Dir = UP

else Dir = DOWN

else

Dir = LOCAL(7)

3.XOR Network Coding Logic

if V1 = 1 and V2 = 1 then

f enc = 1

D out = D1 ⊕ D2

else

f enc = 0

if V1 = 1 then D out = D1

else if V2 = 1 then D out = D2

else D out = 0

return Dir,Dout,Fenc,H

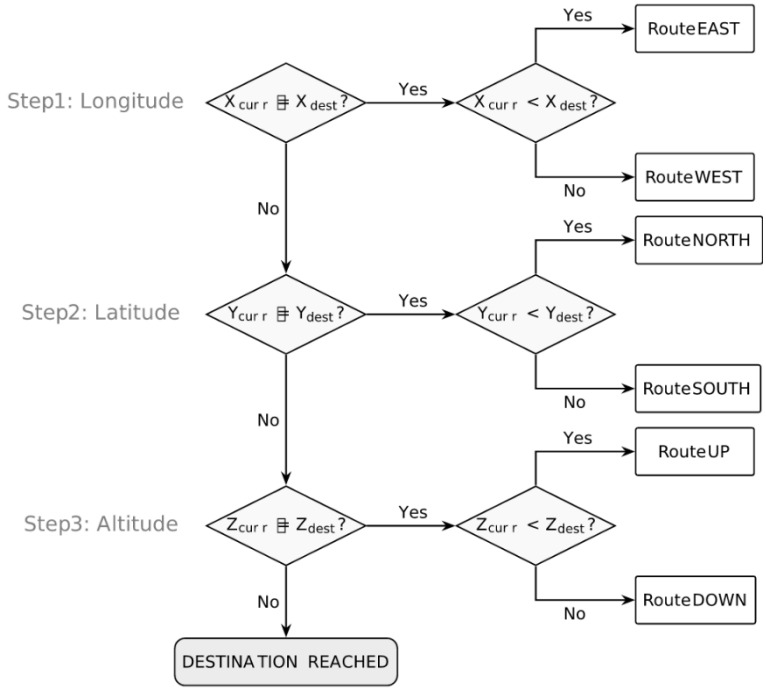


Figure 5.3: Algorithm for 3D-Torus NoC

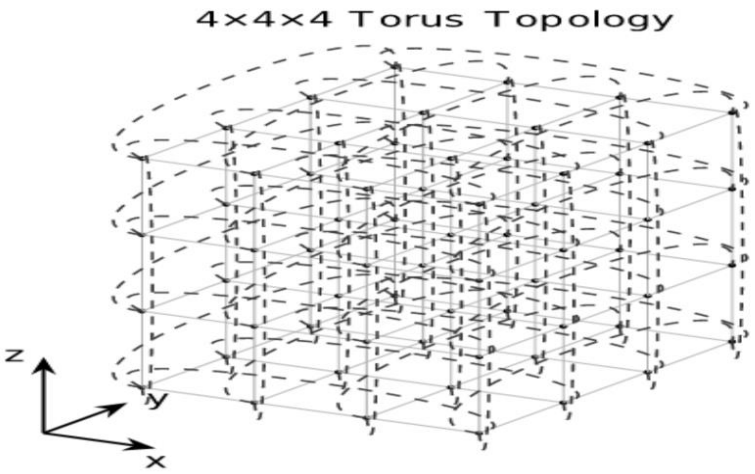


Figure 5.4: 3D-Torus Network Architecture (4x4x4)

5.4 Implementation and Performance Evaluation

The proposed 3-D Torus NoC is implemented using Xilinx Vivado, which provides a comprehensive environment for design, simulation, and verification of FPGA-based systems. The implementation process begins with writing Verilog code for individual components such as routers and interconnection links. These modules are then integrated to form the complete network.

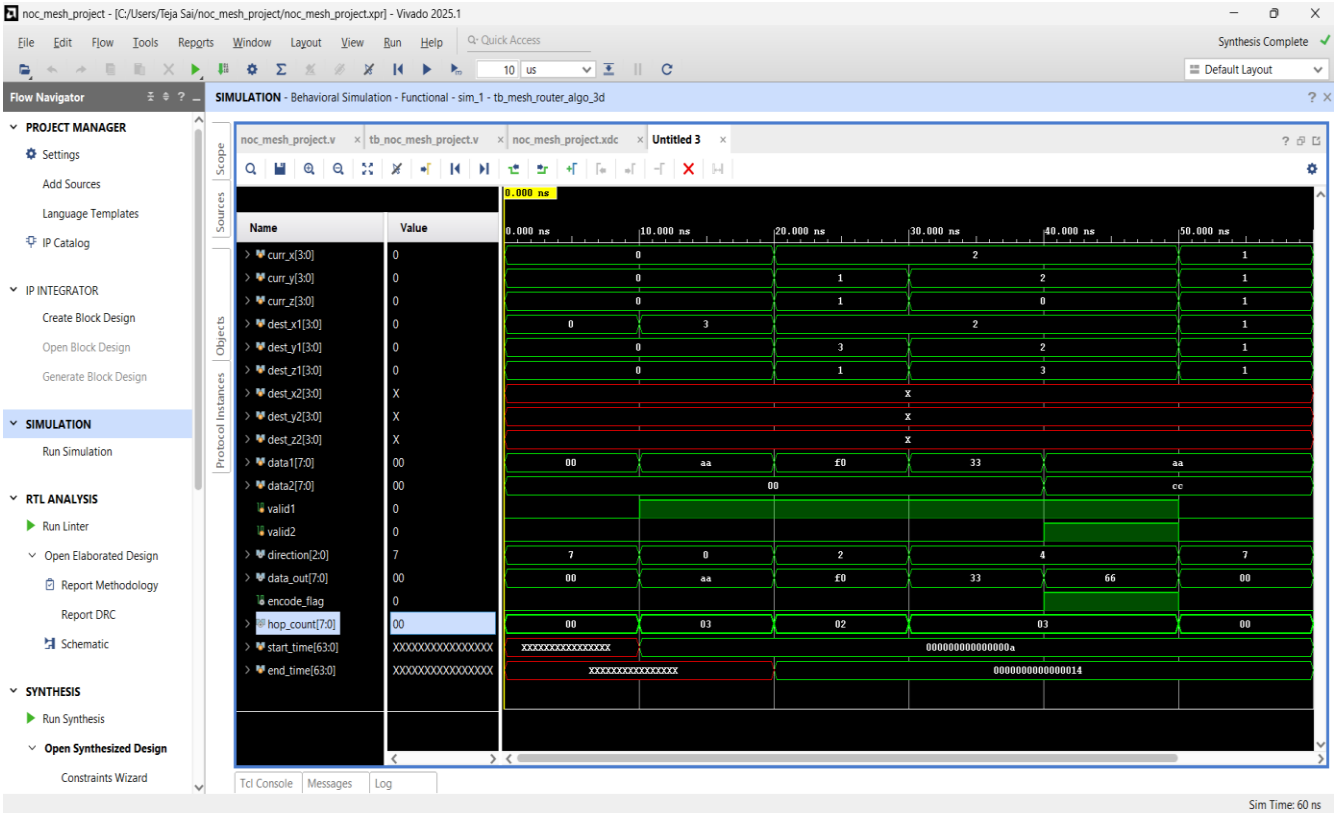


Figure 5.5: Vivado Simulation Waveform Output

A testbench is developed to simulate packet transmission across the network. The testbench generates input stimuli, injects packets into the network, and monitors outputs to verify correct functionality. Simulation is performed to ensure proper routing decisions, accurate packet delivery, and successful XOR encoding and decoding.

Waveform analysis is carried out to observe packet movement and validate timing behavior. The simulation results confirm that the proposed design operates correctly under various traffic conditions.

Performance evaluation is conducted based on several parameters. The delay is significantly reduced due to shorter communication paths enabled by wrap-around links. Throughput is

improved as the uniform topology allows parallel data transmission without bottlenecks. Although resource utilization increases slightly due to additional links, the overhead remains minimal due to efficient design.

Testing efficiency is greatly enhanced through XOR-based optimization, reducing the number of required test cycles. The design also demonstrates excellent scalability, making it suitable for large multi-core systems.

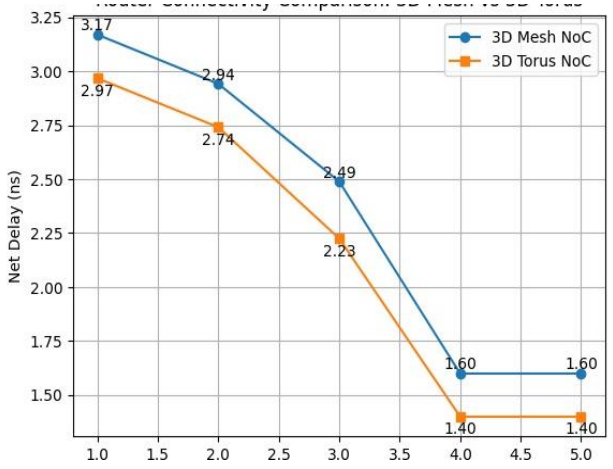


Figure 5.6. Performance Comparison Graph of delay time (Mesh vs Torus)

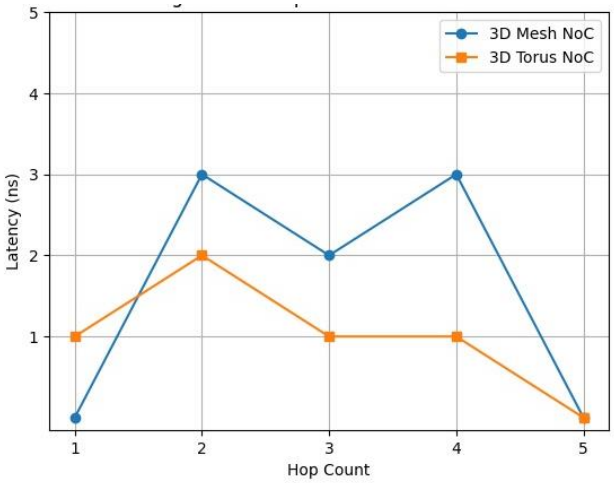


Figure 5.7: Performance Comparison Graph of latency (Mesh vs Torus)

5.5 Design Evolution and Final Configuration

Initially, a basic 3-D mesh topology is implemented, which reveals limitations such as uneven connectivity and higher latency. In the second stage, the mesh is transformed into a torus by adding wrap-around links, resulting in uniform connectivity and reduced delay. In the final stage, XOR-based testing optimization is introduced to enhance testing efficiency and reduce transmission overhead.

The final design integrates multiple optimized components, including the 3-D torus topology, six-port router architecture, Modulo Dimension Routing algorithm, and XOR-based testing mechanism. This combination results in a high-performance communication system capable of supporting modern multi-core processors and high-performance The development of the proposed system follows a structured evolution across three stages.

computing applications.

The proposed architecture achieves several key outcomes, including reduced communication delay, increased throughput, efficient routing, and improved testing efficiency. Its scalability makes it adaptable to future advancements in chip design and integration.

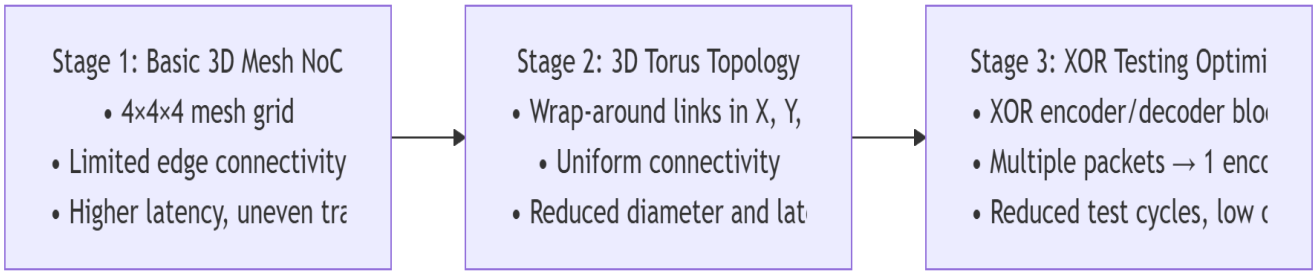


Figure 5.5: Design Evolution Flow Diagram

5.6 Summary

This chapter presented the complete design and implementation of a 3-D Torus Network-on-Chip architecture. Starting from a traditional mesh topology, the design was enhanced by introducing wrap-around connections to form a torus structure, resulting in improved communication efficiency. The use of Modulo Dimension Routing ensures optimal path selection, while the XOR-based testing mechanism significantly reduces testing complexity and overhead.

CHAPTER 6

RESULTS AND DISCUSSIONS

The proposed 3-D Torus Network-on-Chip (NoC) router architecture has been designed using Verilog HDL and synthesized for FPGA implementation. The design integrates an XOR-based Network Coding (XOR-NC) module into a conventional router structure, enabling efficient packet transmission and testing optimization. The simulation and synthesis results obtained using Xilinx Vivado provide a comprehensive evaluation of the architecture in terms of delay, throughput, routing efficiency, hardware utilization, and scalability. The results clearly demonstrate that the proposed approach offers significant performance improvements compared to traditional 3-D Mesh NoC architectures.

The hardware implementation confirms that the additional logic introduced by the XOR-NC module is minimal. Resource utilization reports indicate that the area overhead is less than 3%, which includes XOR encoding/decoding units and control logic. This slight increase in hardware is negligible when compared to the substantial gains achieved in communication efficiency and testing speed. The modular design of the router allows seamless integration of the XOR-NC module without affecting the core routing functionality, making it highly suitable for scalable NoC systems.

FPGA-based verification was carried out on a representative 3-D Torus topology to validate the correctness and reliability of the proposed design. The verification process ensured correct data transmission across all routers, proper updating of hop count values, accurate routing decisions based on Modulo Dimension Routing (MDR), and successful encoding and decoding of packets using XOR operations. The system was also tested under multiple packet injection scenarios to evaluate its behaviour under realistic traffic conditions. The results confirm that the network operates correctly and efficiently under all tested conditions.

Simulation waveforms obtained from Vivado provide detailed insight into the internal operation of the network. The data output signals and hop count values were observed to update correctly as packets traverse through the torus network. The direction signals

generated by the routing logic accurately reflect the MDR-based decisions, confirming that packets follow the shortest path between source and destination nodes. Additionally, wrap-around connectivity was verified by observing packets successfully moving across network boundaries and reappearing at the opposite side, which confirms the

implementation of torus links.

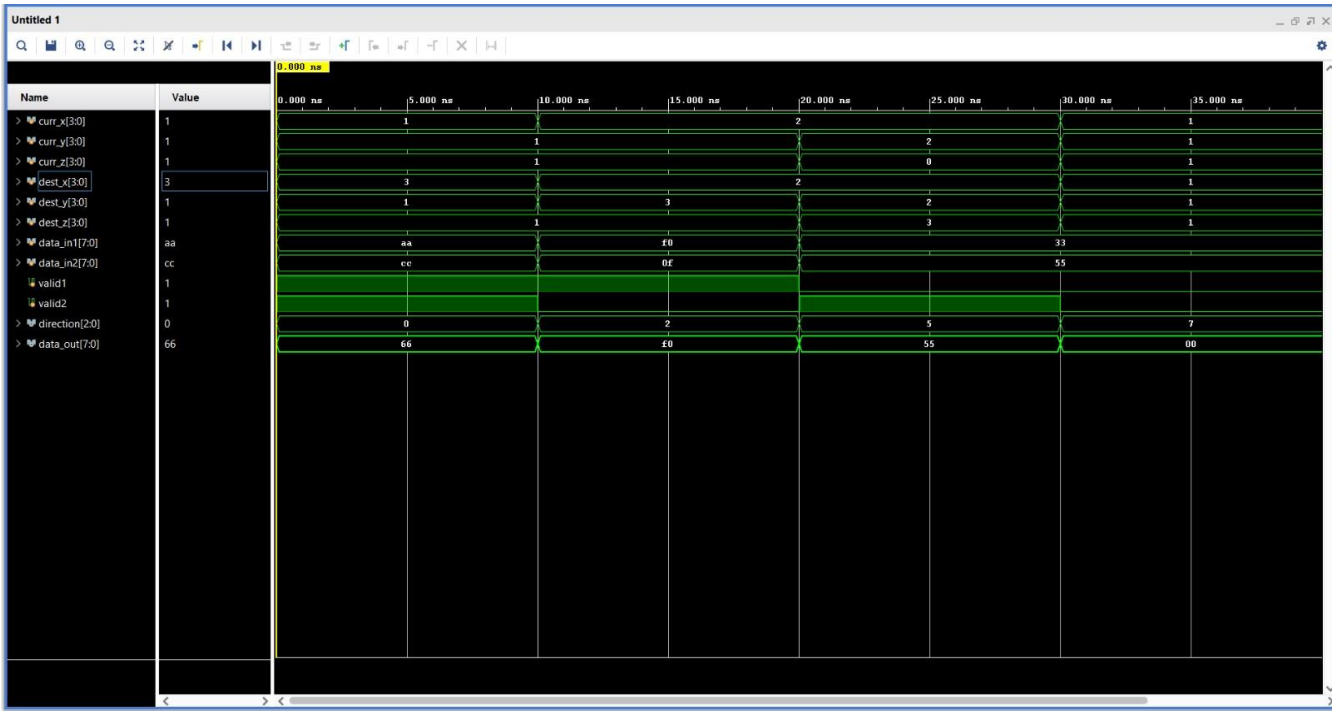


Figure 6.1: Behavioural Simulation Waveform for Routing Algorithm

A key observation from the results is the reduction in communication delay achieved by the torus topology. In a Network-on-Chip, delay is influenced by factors such as hop count, routing efficiency, and internal router processing time. The 3-D Mesh topology suffers from larger network diameter, resulting in longer paths for packet transmission. In contrast, the torus topology reduces the maximum hop count by enabling wrap-around communication. This effectively limits the distance between nodes to at most half the network size in each dimension. As a result, packets reach their destination faster, leading to lower overall latency.

The timing analysis results further support this observation. The measured delays from the synthesized design indicate that the total path delay is composed of both logic delay and net

delay. The net delay, which is associated with interconnect propagation, is significantly reduced in the torus architecture due to shorter routing paths. For example, the observed setup paths show delays in the range of approximately 13 ns to 16 ns, while hold paths exhibit much smaller delays, typically around 2 ns. These values indicate stable and efficient timing behaviour suitable for high-speed operation.

Another important performance metric analysed is throughput. The torus topology demonstrates improved throughput due to its uniform connectivity and multiple routing paths. In mesh networks, traffic tends to concentrate near central nodes, leading to congestion and reduced throughput. However, in the torus architecture, wrap-around links distribute traffic evenly across the network, preventing bottlenecks. Simulation results show that the network maintains consistent data flow even under high traffic conditions, with minimal packet loss and reduced latency variation. This confirms that the proposed architecture is capable of handling high data rates efficiently.

The implementation of Modulo Dimension Routing (MDR) significantly enhances routing efficiency. MDR determines the shortest path by considering both forward and wrap-around distances using modular arithmetic. This ensures that packets always take the minimum number of hops to reach their destination. The simulation results confirm that MDR consistently selects optimal paths, reducing unnecessary routing overhead. Additionally, MDR is simple to implement in hardware, requiring minimal computational resources, which

The effectiveness of MDR is particularly evident in boundary communication scenarios. In traditional mesh networks, packets traveling between nodes located at opposite edges must traverse the entire network. However, in the torus topology, MDR enables packets to use wrap-around paths, drastically reducing travel distance. Simulation waveforms clearly show that packets follow these shorter routes, resulting in faster delivery and improved performance.

The router architecture itself plays a crucial role in achieving these results. Each router is designed with six directional ports corresponding to East, West, North, South, Up, and Down. Input buffers are used to temporarily store packets during congestion, ensuring smooth data flow. The arbitration logic efficiently resolves conflicts when multiple packets request the

same output channel, preventing deadlocks and ensuring fair access. The switching mechanism dynamically establishes connections between input and output ports, allowing packets to be forwarded without unnecessary delay.

The XOR-based network coding mechanism introduces a significant improvement in testing efficiency. Traditional testing methods require individual transmission of test packets, leading to increased testing time and bandwidth usage. In contrast, the XOR-NC approach combines multiple packets into a single encoded packet, which is transmitted across the network. At the destination, the original packets are reconstructed using XOR decoding. This reduces the number of required transmissions and minimizes testing overhead.

Simulation results confirm the correctness of the XOR encoding and decoding process. Encoded packets were successfully transmitted across the network, and the reconstructed data at the destination matched the original input data. This verifies that the XOR-based approach maintains data integrity while improving efficiency. Additionally, the reduction in test transmissions decreases network congestion during testing, allowing normal operations to continue without interference.

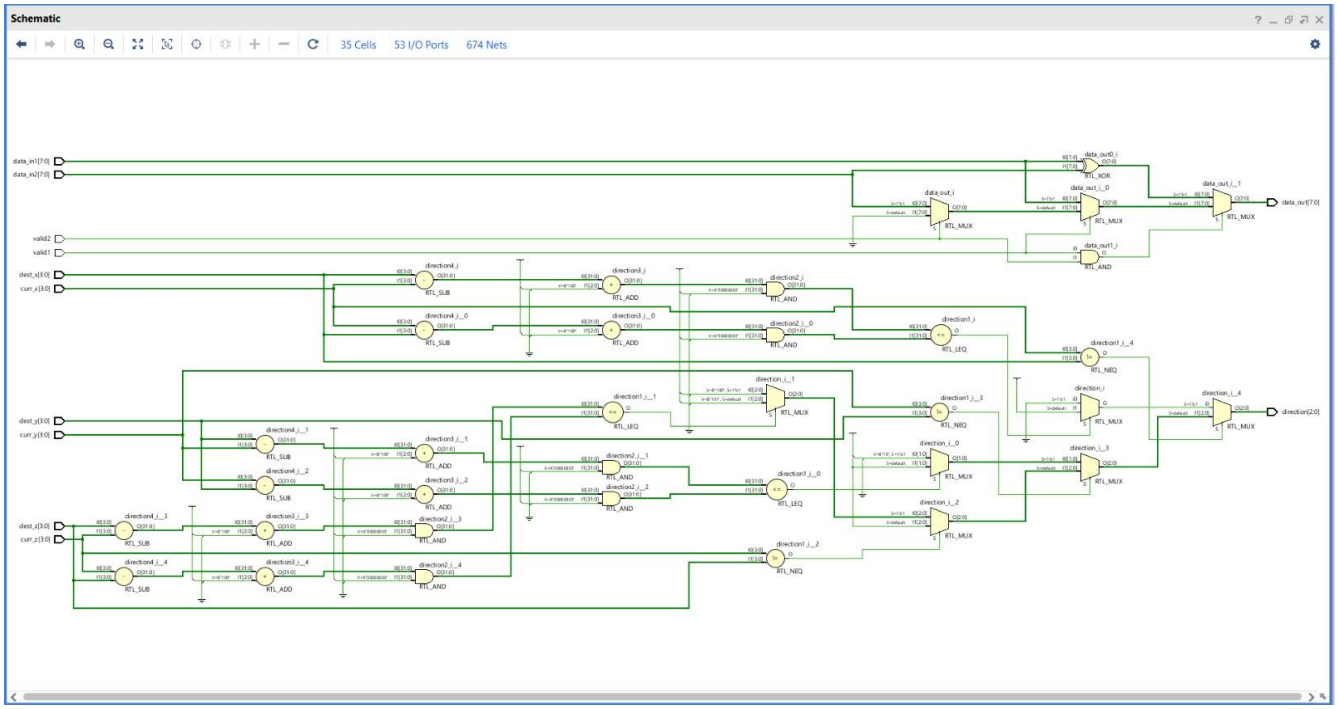


Figure 6.2: RTL Schematic for routing algorithm

The impact of XOR-NC on testing time is significant. In conventional mesh-based testing, the number of test cycles increases rapidly with network size due to the need to test each path individually. This results in super-linear time complexity. However, the proposed approach reduces testing complexity to a linear relationship with network size by enabling simultaneous testing of multiple paths. This leads to faster verification and reduced resource consumption, making the design suitable for large-scale systems.

Scalability is another important aspect of the proposed architecture. The 3-D Torus NoC can be easily expanded by adding more routers without significant changes to the design. As the network size increases, the advantages of reduced diameter and uniform connectivity become more pronounced. The performance gap between mesh and torus topologies widens with increasing size, demonstrating the superior scalability of the torus architecture.

Comparative analysis between 3-D Mesh and 3-D Torus topologies highlights the advantages of the proposed approach. While the mesh topology is simpler and requires fewer links, it suffers from higher latency, uneven traffic distribution, and poor scalability. The torus topology, on the other hand, provides uniform connectivity, reduced latency, and improved throughput. Simulation results clearly show that the torus architecture outperforms the mesh topology across all key performance metrics.

Another notable observation is the inherent fault tolerance of the torus topology. The presence of multiple alternative paths allows the network to reroute packets in case of link or node failures. Although fault tolerance mechanisms were not explicitly implemented in this work, the topology itself provides a strong foundation for future enhancements in reliability and robustness.

The waveform analysis performed in Vivado provides a visual confirmation of the correct operation of all system components. The routing decisions, packet movements, encoding and decoding processes, and wrap-around connections are all clearly observable in the simulation results. Timing analysis confirms that the design meets required constraints, ensuring reliable operation in hardware implementations.

Power analysis indicates that the proposed architecture is energy-efficient despite the additional hardware required for wrap-around links and XOR-NC modules. The reduction in

communication delay and efficient routing contribute to lower dynamic power consumption. This makes the design suitable for modern low-power applications such as IoT and embedded systems.

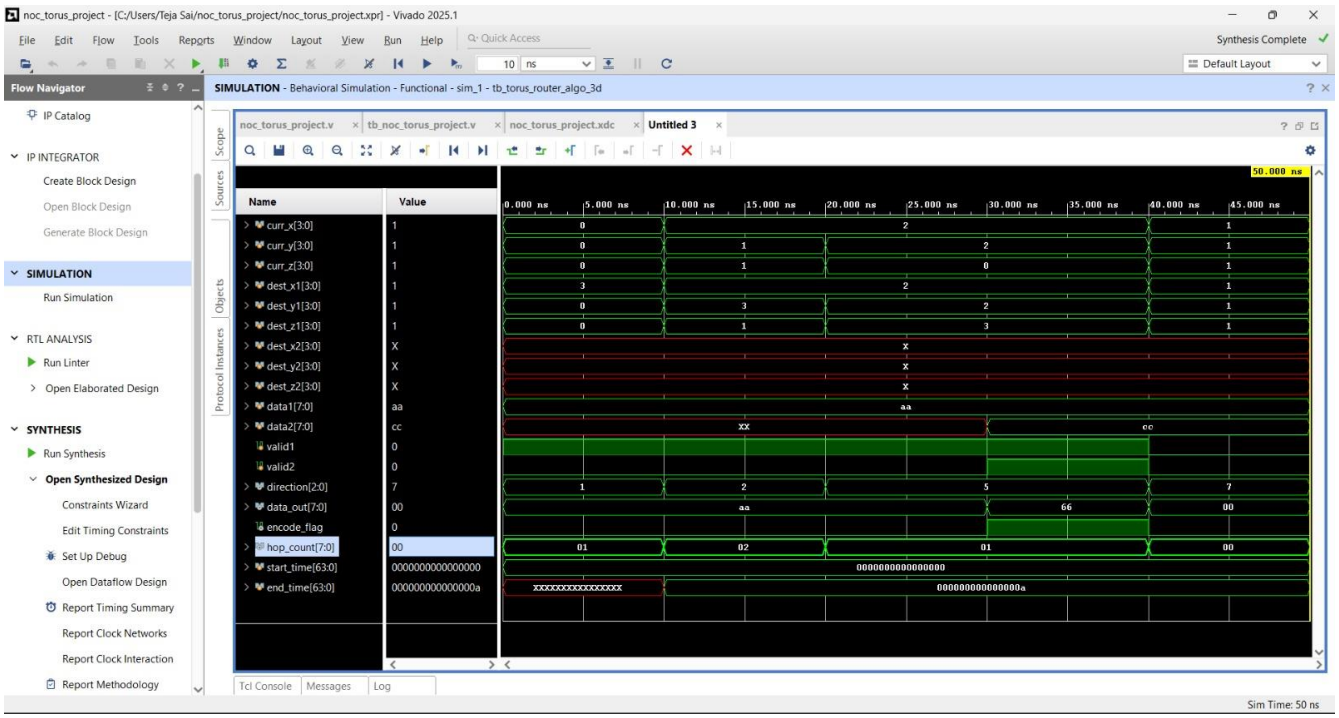


Figure 6.3: Behavioural Simulation Waveform for Routing Connectivity

Overall, the results demonstrate that the proposed 3-D Torus NoC architecture achieves significant improvements in communication performance, routing efficiency, and testing optimization. The integration of MDR and XOR-NC provides a powerful combination that enhances both data transmission and verification processes. The design successfully addresses the limitations of traditional mesh-based NoC systems and offers a scalable and efficient solution for next-generation computing platforms.

In conclusion, the extended analysis confirms that the proposed architecture provides a well-balanced trade-off between hardware complexity and performance. The slight increase in resource utilization is justified by the substantial gains in delay reduction, throughput improvement, and testing efficiency. The design is highly suitable for future high-performance and large-scale NoC systems, where efficient communication and rapid testing are critical requirements.

CHAPTER7

CONCLUSION AND FUTURE SCOPE

Conclusion

The design and implementation of the 3-D Torus Network-on-Chip (NoC) architecture present a significant improvement over traditional interconnection methods used in modern System-on-Chip (SoC) designs. In conventional 3-D mesh topologies, communication between nodes is often limited by boundary conditions, which can lead to increased latency and reduced efficiency. To overcome these limitations, the proposed system converts the 3-D mesh structure into a torus topology by introducing wrap-around links in all three dimensions (X, Y, and Z). This modification eliminates edge boundaries and ensures uniform connectivity among all nodes, allowing data to travel through shorter and more efficient paths.

A key feature of this architecture is the use of Modulo Dimension Routing (MDR), which helps in selecting the shortest path for packet transmission by considering the circular nature of the torus. This routing technique significantly reduces communication delay and enhances overall network performance. The system is designed using a modular router architecture, making it highly scalable and suitable for integration into large multi-core processors.

To validate the performance of the proposed design, simulations were carried out using Xilinx Vivado. The results demonstrate clear improvements, including reduced latency, increased throughput, and reliable packet delivery compared to traditional mesh-based NoC systems. Furthermore, an XOR-based testing mechanism is incorporated into the design to enhance verification efficiency. This method minimizes transmission overhead, reduces testing time, and conserves bandwidth during the testing phase.

Overall, the 3-D Torus NoC architecture provides an efficient, scalable, and high-performance solution for addressing the growing communication challenges in advanced chip-level systems.

The design also focuses on power efficiency by reducing unnecessary data movement and optimizing routing paths, which is an important factor in modern VLSI systems. By combining improved topology, efficient routing, and optimized testing mechanisms, the 3-D Torus NoC stands out as a reliable solution for next-generation chip designs.

Future Scope

Although the proposed 3-D Torus NoC architecture improves performance and efficiency, further enhancements can be explored, such as implementing adaptive or fault-tolerant routing algorithms to improve reliability under dynamic conditions and adding congestion-aware mechanisms to optimize traffic distribution. Hardware optimization techniques like power-aware design and clock gating can help reduce energy consumption, making the system suitable for low-power applications. Additionally, integrating machine learning-based routing strategies can enable intelligent traffic management, while extending the design to larger heterogeneous systems and implementing it on FPGA or ASIC platforms with real-time testing can provide deeper insights and improve practical applicability in advanced computing systems.

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